Yue-Hong Chen

Ph.D. Candidate in Astronomy

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Research Interests

My research bridges solar and stellar physics. Starting from observational studies of solar eruptive activity, I have extended my work to high-resolution magnetohydrodynamic (MHD) simulations of Sun-like stars, covering the full chain from stellar coronae and winds to coronal mass ejections and the associated energetic particle events.

Education

Sep 2023 – Present **Ph.D. in Astronomy**, *Nanjing University*, Nanjing, China.
Advisor: Prof. Xin Cheng.
Research focus: high-resolution MHD modelling of stellar coronae, winds, and CMEs in Sun-like stars.

Sep 2025 – Present **Visiting Ph.D. Researcher**, *Leibniz Institute for Astrophysics Potsdam (AIP)*, Potsdam, Germany.
Host: Dr. Julián D. Alvarado-Gómez.
Research focus: stellar wind and CME modelling of Sun-like stars at varying rotation rates.

Sep 2021 – Jun 2023 **M.Sc. in Astronomy**, *Nanjing University*, Nanjing, China.
Advisor: Assoc. Prof. Yu Dai.
Research focus: observational study of solar flares and filament eruptions.

Additional Research Visits

Sep – Nov 2024 **Visiting Researcher**, *Leibniz Institute for Astrophysics Potsdam (AIP)*, Potsdam, Germany.

Aug 2023 **Joint-Training Visiting Student**, *Yunnan University*, Kunming, China.
Host: Prof. Chun Xia. Research focus: MPI-AMRVAC modelling.

Publications

First-Author Refereed Articles

2026 **Chen, Y.-H.**, Alvarado-Gómez, J. D., Cheng, X., See, V., Dai, Y., Korpi-Lagg, M. J., Warnecke, J., Xing, C., & Ding, M. *High-Resolution Modeling of Coronae and Winds in Solar-type Stars with Varying Rotation Rates. II. Stellar Winds*. *ApJ*, 1011, 49. doi:10.3847/1538-4357/aeab7e; arXiv:2609.28067

2025 **Chen, Y.-H.**, Alvarado-Gómez, J. D., Cheng, X., Dai, Y., Shi, T., Poppenhäger, K., Xing, C., Inoue, S., Warnecke, J., Korpi-Lagg, M. J., & Ding, M. *High-Resolution Modeling of Coronae and Winds in Solar-Type Stars with Varying Rotation Rates. I. X-Ray Coronae*. *ApJ*, 995, 83. doi:10.3847/1538-4357/ae1697

2023 **Chen, Y.**, Cheng, X., Chen, J., Dai, Y., & Ding, M. *Observations of a Failed Solar Filament Eruption Involving External Reconnection*. *ApJ*, 959, 67. doi:10.3847/1538-4357/ad09d8

2023 **Chen, Y.**, Dai, Y., & Ding, M. *An Atypical Plateau-Like Extreme-Ultraviolet Late-Phase Solar Flare Driven by the Nonradial Eruption of a Magnetic Flux Rope*. *A&A*, 675, A147. doi:10.1051/0004-6361/202345914

Co-Authored Refereed Articles

2026 Sun, Z., Alvarado-Gómez, J. D., Tian, H., Cohen, O., Drake, J. J., Poppenhäger, K., & **Chen, Y.-H.** *Intense but Harmless: Exo-Space Weather around an M Dwarf with a Single-hemisphere Dynamo*. *ApJ*, 1009, 97. doi:10.3847/1538-4357/ae976d; arXiv:2608.11087

2026 Xie, T. Y., Zhao, Z. H., Cheng, X., **Chen, Y.-H.**, Zheng, Z., Hao, Q., Li, C., & Ding, M. D. *Statistics of Solar Filament Mass Based on CHASE Sun-as-a-star Spectroscopic Observations*. *ApJL*, 1008, L45. doi:10.3847/2041-8213/ae9850; arXiv:2608.11826

2025 Zhou, P., Mao, J., Zhang, L., et al. (incl. **Chen, Y.-H.**; eXTP Collaboration). *Observatory Science with eXTP*. *Sci. China Phys. Mech. Astron.*, 68, 119507. doi:10.1007/s11433-025-2799-0; arXiv:2506.08367

In Preparation

Chen, Y.-H., et al., *Super-CMEs, Solar-like Proton Events: Global 3D MHD Modeling of Shock-Accelerated Energetic Particles on the Young Sun EK Draconis*, in prep.

Conference Talks (Oral)

All entries below are oral presentations given as presenting author.

Nov 2026 IAU Symposium 410, "New Look at Solar and Stellar Coronae." Shenzhen, China. (Scheduled, Nov 29 – Dec 4.)

Nov 2026 "Sun–Earth Connection through Stellar Analogs" Workshop. University of Graz, Austria. Contributor. (Scheduled.)

Aug 2026 COSPAR 2026, Florence, Italy.

Aug 2026 Magnetic Eclipse, Valencia, Spain.

Jun 2026 Cool Stars 23, Tokyo, Japan. *(Awarded Best Poster, granted oral presentation opportunity.)*

Mar 2026 89th DPG Annual Meeting & SMuK Spring Conference, Extraterrestrial Physics (EP) Division. FAU Erlangen-Nürnberg, Germany.

Aug 2025 16th Zhang Heng Symposium / Joint Solar–Stellar Physics Workshop (Centenary of Chinese Hand-Drawn Sunspot Records). Qingdao, China.

Jul 2025 IAU Symposium 400, "Solar and Stellar Multi-Scale Activity." EAFIT University, Medellín, Colombia.

May 2025 "Stellar X-Ray Flares in the Einstein Probe Era" Workshop (EP Science Center & NAOC Solar Magnetic Activity Group). Beijing, China.

Apr 2025 2nd International Conference of Deep Space Sciences (ICDSS 2025). Hefei, China. *(Best Presentation Award.)*

Jul 2024 45th COSPAR Scientific Assembly. BEXCO, Busan, Korea.

May 2024 IAU Symposium 388, "Solar and Stellar Coronal Mass Ejections." Jagiellonian University Observatory, Kraków, Poland.

2023 Annual Meeting of the Chinese Solar Physics Community. Chengdu, China.

Seminars & Talks

Nov 2026 Konkoly Observatory. Budapest, Hungary. (Scheduled, Nov 19.)

Oct 2026 University of Tübingen. Tübingen, Germany.

Oct 2026 KU Leuven. Leuven, Belgium.

Oct 2026 Leiden Observatory, Leiden University. Leiden, the Netherlands.

Sep 2026 Max Planck Institute for Solar System Research (MPS). Göttingen, Germany.

Sep 2026 ASTRON, Netherlands Institute for Radio Astronomy. Dwingeloo, the Netherlands.

Nov 2024 Leibniz Institute for Astrophysics Potsdam (AIP). Potsdam, Germany.

Workshop Organisation

Dec 2026 *Co-organiser*, mini-workshop "Beyond the Sun: Stellar Eruptions and Planetary Environments." Nanjing University, China. (Scheduled.)

Grants & Funding

Apr 2024 – May 2026 **Formation and Evolution of Stellar Coronal Mass Ejections**. *Principal Investigator*. Postgraduate Research & Practice Innovation Program, awarded by the Ministry of Education of Jiangsu Province, China (EUR 4,000).

Sep 2024 – Aug 2027 **Lower-Atmospheric Signatures of Coronal Mass Ejection Initiation**. *Co-Investigator* (numerical simulation). Natural Science Foundation of Jiangsu Province (Youth Program). PI: Dr. Chen Xing.

Honors & Awards

2026 **Chenxi Yangfan Scholarship**, Nanjing University, China (EUR 1,300).

2025 **Joint Ph.D. Scholarship**, China Scholarship Council (EUR 20,250). *Visiting Researcher* at the Leibniz Institute for Astrophysics Potsdam (AIP), Germany, Sep 2025 – Nov 2026, on "Modelling Coronal Mass Ejections on Solar-type Stars".

2024 **Tiangong International Overseas Exchange Scholarship**, Nanjing University (EUR 5,550). *Visiting Researcher* at the Leibniz Institute for Astrophysics Potsdam (AIP), Germany, Sep – Nov 2024, on "Modelling Stellar Coronae and Winds from Solar-type Stars".

Teaching & Supervision

TA *Introduction to Space Physics* (undergraduate course), Nanjing University.

2023 Co-supervised two undergraduate students on an early-stage research project analysing Sun-as-a-star coronal dimming spectra.

Languages

Mandarin Chinese — Native

English — Working proficiency

References

Ph.D. advisor — Prof. Xin Cheng, School of Astronomy and Space Science, Nanjing University.

Visit host — Dr. Julián D. Alvarado-Gómez, Leibniz Institute for Astrophysics Potsdam (AIP).

M.Sc. advisor — Assoc. Prof. Yu Dai, School of Astronomy and Space Science, Nanjing University.